

2(a)(i)	distance from GM rice plants and direction from GM rice plants ;	1
2(a)(ii)	apply herbicide to young plants ; count the number of young plants that survive ;	2
2(a)(iii)	any two from: 1 as distance from GM rice plants increases, (percentage) gene flow decreases / pollen dispersal decreases ; ora 2 (percentage) gene flow / distance of pollen dispersal, is higher in direction of (normal) wind / NW direction (than other directions) ; ora 3 pollen cannot disperse further than 10 m / no gene flow at 10m (or further) ;	2
2(b)(i)	0.036 ;	1
2(b)(ii)	any three from: 1 critical value (for $p = 0.05$) = 2.145 or critical value (for $p = 0.01$) = 2.977 ; 2 value for $t / 9.043$ / calculated value, is greater than critical value ; 3 null hypothesis is rejected ; 4 there is a <u>significant</u> difference between percentage gene flow from GM rice plants to weedy rice plants and percentage gene flow from weedy rice plants to GM rice plants ;	3

Question	Answer	Marks
2(b)(iii)	any one from: 1 investigation only, carried out once / not repeated ; 2 investigation only on, wind-pollinated crop / one crop species / (GM and weedy) rice / one non-crop species ; 3 investigation only investigated, one / two, gene(s) ; 4 investigation only, in one geographical area ; 5 investigation only in, experimental trials / without herbicide use ; 6 no long-term data / no data over several generations ; 7 gene flow (from GM plants to wild plants), was shown to occur ; 8 <i>idea that</i> wild plants with herbicide-resistant gene, difficult to remove / have selective advantage ;	1